PACKET SNIFFING USING RAW SOCKETS

Aim:

To implement a packet sniffing using python raw sockets and display the source IP, destination IP & protocol of captured packets.

Code:-

```
import socket, struct
sock = socket.socket (
        socket.AF_INET,
        socket.SOCK_RAW,
        socket.IPPROTO_IP
    )
print ("Packet Sniffer started ...")
print ("Waiting for packets ... \n")
while True:
    packet, address = sock.recvfrom (65536)
    ip_header = packet [0:20]
    iph = struct.unpack (
        "!BBHHH BBH 4s4s",
        ip_header)
    source_ip = socket.inet_ntoa (iph[8])
    destination_ip = socket.inet_ntoa (iph[9])
    protocol = iph[6]
    print (" Source IP:", source_ip)
    print (" Destination IP:", destination_ip)
    print (" Protocol:", protocol)
    print ("Packet size:", len (packet), "bytes")
```

SAMPLE OUTPUT:

packet sniffer started...
waiting for packets...

Source IP: 192.168.1.10
Destination IP: 142.250.183.14
Protocol: 6
Packet size: 60 bytes
Source IP: 8.8.8.8
Destination IP: 192.168.1.10
Protocol: 1
Packet size: 84 bytes

